SPECTRUM

TIME: 1 hr **MARKS: 40**

SECTION I

[*All* questions are compulsory]

Question 1: **[5]**

[i] The angle through which a light of a particular colour can deviate when incident on a optical glass prism is called…….

[a] angle of incidence

[b] angle of reflection

[c] critical angle

[d] angle of refraction

[ii] Rock salt prism does not absorb ……. radiations.

[a] radio waves

[b] microwaves

[c] nano waves

[d] infra-red

[iii] Refrangibility of ……. light is least and ….. light is maximum.

[a] red, violet

[b] cyan magenta

[c] violet red

[d] black, white

[iv] ..... rays are used for sterilizing surgical instruments.

[a] Infra-red

[b] blue

[c] ultraviolet

[d] red

[v] ..... rays bring about chemical reactions in silver salts.

[a] Violet

[b] indigo

[c] orange

[d] ultraviolet

Question 2: **[5]**

[i] In the atmosphere scattering is maximum for ..... [radio waves / microwaves]

[ii] The splitting of polychromatic light into its constituent colours is called .....[dispersion / spectrum]

[iii] The rays used to determine cracks in the skull of a brain are .... [Infra-red rays / X rays]

[iv] The machine used to determine the malignant cells in our body shown in the figure is ..... [ECG machine / MRI machine]



[v] The radiations used to take such photographs are ……. [red / infrared]



Question 3:

[a] Give two sources of producing ultraviolet radiations? [2]

Ans: Two sources of ultraviolet radiations are: [i] Sun and [ii] Electric arc lamp.

[b] How can the presence of infra-red radiations be detected? [2]

Ans: In the presence of infra-red radiations the temperature recorded by a blackened bulb thermometer rises.

[c] Define the phenomenon of dispersion. [2]

Ans: The splitting up of polychromatic ray of light into its constituent colours when passed through an optical glass prism is called as dispersion.

[d] State any two properties of infra-red radiations which differ from visible light. [2]

Ans: Two properties of infra-red radiations which differ from visible light are:

[i] wavelength of infra-red radiations are longer than visible light

[ii] Infra-red radiations have more heating effect than visible light

[e] Photographic dark rooms provided with infra-red lamps. Why? [2]

Ans: Infra-red radiations do not affect normal photographic films so used in photographic dark rooms.

SECTION II

[Attempt *any one* question out of two]

Question 4:

[a] Give reasons for the following: [4]

[i] Rock salt prism is used instead of glass prism to obtain infrared spectrum.

Ans: Glass prism absorbs infra-red while rock salt does not absorb them.

[ii] Quartz prism is used to instead of glass prism to obtain ultraviolet spectrum

Ans: Glass prism absorbs ultraviolet radiations while quartz prism does not absorb them.

[b] State three properties of ultraviolet radiations similar to visible light. [3]

Ans: The three properties of ultraviolet radiations similar to visible light are:

[i] They travel at the speed of 3 x 108ms-1 in vacuum

[ii] both the waves are electromagnetic waves

[iii] both the waves obey the laws of reflection and refraction.

[c] State the range of wavelength for: [i] Infra-red radiations [ii] visible light and [iii] ultraviolet radiations. [3]

Ans:

[i]The range of wavelength for infra-red spectrum is 8000°A to 10^{7}°A

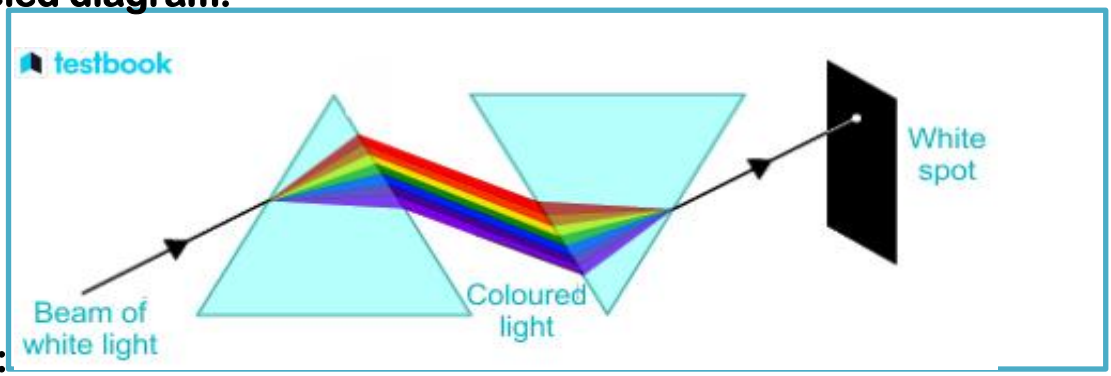
[ii] the range of wavelength for visible light is 4000°A to 8000°A

[iii] Range of wavelength for spectrum of ultraviolet is 100°A to 4000°A

Question 5:

[a] Explain the newton's biprism experiment with the help of suitable labeled diagram. [4]

Ans:



When polychromatic light such as sunlight is incident on the first prism called the dispersing prism it splits up into its constituent colours due to different wavelengths till it refracts on the emerging surface of the first prism. These emergent rays are made to incident on an inverted glass prism such that these recombine together while traveling through the second prism and emerge as a single beam of white light which is parallel to the original incident light.

[b] Calculate the wavelength of the electromagnetic wave whose frequency is 10^{12}Hz. Name the given electromagnetic wave. [3]

Ans: Since the wave is an electromagnetic wave the velocity of the wave is $3 \times 10^{8} ms^{-1}$

Using the relationship,

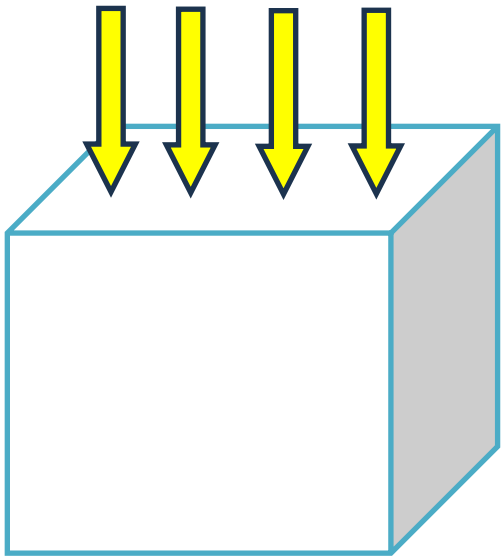
$$\text{Wave velocity} = \text{frequency} \times \text{wavelength.}$$

$$3 \times 10^{8} = 10^{12} \times \lambda$$

$$\lambda = 3 \times 10^{-4} \text{ m}$$

These waves are categorized as infra-red radiations.

[c] A transparent cuboid is illuminated from the top surface by means of sunlight made of a material such that it reflects red light scatters green light, transmits blue light, and absorbs yellow, orange, indigo and violet light. Which colour would this cube appear from: [i] top [ii] below [iii] lateral side surfaces? [3]



Ans:

When viewed from above the colour of surface would appear red

When viewed from below the colour of appearance would be blue

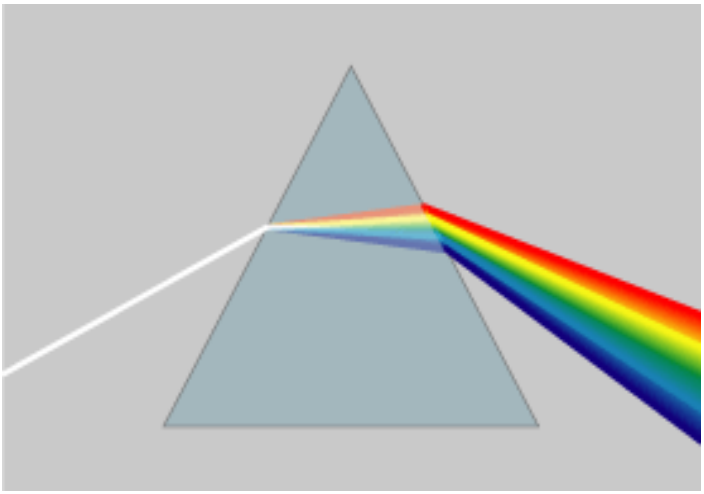
When viewed from lateral surfaces the appearance would be green.

Question 6:

a] Explain by means of a diagram cause of dispersion of polychromatic light through a prism. [3]

Ans: Dispersion of light is the phenomenon of splitting up of polychromatic light when incident on an optical glass prism. It occurs because polychromatic light means light having different wavelengths and when these different wavelengths enter the optical glass prism deviate through different angles travelling at various speeds while going through the prism as shown in the figure below

PHENOMENON OF DISPERSION.



b] In the visible spectrum of white light state the colour that has the longest wavelength and the colour which has the shortest wavelength. How does this affect the scattering effect. [3]

Ans: In visible spectrum of white light the colour of longest wavelength is red and hence scatters the least while violet has the shortest wavelength and hence scatters maximum as scattering is inversely proportional to fourth power of wavelength.

[c] Enlist the complete electromagnetic spectrum in increasing order of wavelength. [4]

Ans: The electromagnetic spectrum in increasing order of wavelength is as follows:1. Gamma rays 2. X rays 3. Ultraviolet rays 4. Visible light 5. Infra-red waves 6. Microwaves 7. radio waves.